Department of Electrical and Electronics Engineering

Academic Year 2023 – 2024 Even Semester

Degree, Semester & Branch: IV Semester B.E EEE
Course Code & Title: EE3402 – Linear Integrated circuits
Name of the Faculty member (s): Dr. K. Karthikeyan

Innovative Practice Description

- **Unit / Topic: Unit I / Fabrication of FETs**
- **Course Outcome:** CO 1
- **Topic Learning Outcome:** TLO 3
- **Activity Chosen: Strip sequence**
- **Justification:**
  - ✓ The **Strip Sequence** activity is well-suited for teaching the fabrication process of Field-Effect Transistors (FETs) as it breaks down the complex, multi-step procedure into manageable parts.
  - ✓ The Strip Sequence activity provides a highly visual and engaging way to understand the complex and multi-step process of FET fabrication.
  - ✓ By allowing students to sequentially simulate the fabrication steps, this activity fosters a deeper understanding of each stage in the FET manufacturing process, reinforcing both theoretical and practical aspects of semiconductor fabrication.
- **Time Allotted for the Activity:** 10 minutes
- **Details of the Implementation:**
  - ✓ A set of strips, each representing a key step in the FET fabrication process (e.g., oxidation, photolithography, doping, metallization) has been prepared.
  - ✓ The FET fabrication process and its key steps has been briefly explained to the students.
  - ✓ The strips have been distributed to student groups and I instructed them to arrange them in the correct chronological order to represent the fabrication sequence.
  - ✓ The students have presented their sequences, explained their reasoning.
  - ✓ I have provided feedback on the accuracy of the sequences and addressed any common misconceptions.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO12
CO2	2	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO9**	**PSO1**
	1	**1**
Justification for correlation	The students are working individual and in a team. Hence it is mapped with 1.	The students are understanding the electronic component manufacturing. So, it is mapped with 1.

- **Images / Screenshot of the practice:**

- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***
    - ✓ Students found the Strip Sequence activity effective for visualizing the fabrication stages and connecting theory to practice.
    - ✓ The hands-on nature of the activity helped them understand the intricacies of FET manufacturing, which would have been difficult through theoretical study alone.
    - ✓ Some students noted that the activity improved their retention of the material by allowing them to see and physically manipulate the components.
    - ✓ Some students noted that the complexity of certain steps, such as photolithography, was challenging but rewarding once understood.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
  - ✓ Students gained practical experience by simulating real-world fabrication steps.
  - ✓ The activity facilitated a clearer understanding of how each step in FET fabrication affects the overall process and final device performance.
  - ✓ It helps students grasp the order of operations and the importance of each step in the manufacturing process.
  - ✓ Students actively participate by arranging the strips in the correct sequence, making the learning process more interactive and memorable.
  - ✓ Working in pairs or groups fosters teamwork and collaborative learning.

- ❖ ***Challenges faced in implementation:***
  - ✓ Ensuring sufficient time for students to complete the activity and engage in discussion.
  - ✓ Some students struggled to navigate the simulation software, which required additional guidance and slowed progress.
  - ✓ Limited time made it difficult for students to complete all fabrication steps thoroughly, especially in groups with varying levels of familiarity with the concepts.
  - ✓ Differences in students' prior knowledge and hands-on skills can lead to varied levels of understanding and participation.
  - ✓ Ensuring all students are actively engaged and following the correct sequence can be challenging, requiring effective classroom management.

References:

- ❖ David A. Bell, ‘Op-amp & Linear ICs’, Oxford, Third Edition, 2011
- ❖ D. Roy Choudhary, Sheil B. Jani, ‘Linear Integrated Circuits’, New Age, Fourth Edition, 2018.
- ❖ https://www.universitywafer.com/field-effect-transistor.html
- ❖ https://www.ossila.com/pages/organic-thin-film-field-effect-transistor-fabrication-guide

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Innovative Practice Description

- **Unit / Topic: Unit III / Inverting and Non-inverting amplifier**
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** TLO 6
- **Activity Chosen: Lab Demonstrations with simulation**
- **Justification:**
  - ✓ The **Simulation Activity** is ideal for the topic "Inverting and Non-Inverting Amplifiers" because it provides a hands-on approach to understanding circuit behavior.
  - ✓ By simulating amplifiers, students can visualize input-output relationships, explore real-time voltage gain adjustments, and test circuit designs without physical components, fostering conceptual clarity and practical problem-solving skills.
- **Time Allotted for the Activity:** 25 minutes
- **Details of the Implementation:**
  - ✓ The amplifier configurations, gain, and phase relationships are explained to the students.
  - ✓ Students use simulation software (e.g., Multisim) to build circuits based on provided diagrams.
  - ✓ I provided basic circuit schematics for inverting and non-inverting amplifiers with the students and I encouraged them to modify these circuits (e.g., change resistor values, add components) to explore different configurations.
  - ✓ Students have recorded their simulation results (e.g., voltage readings, waveforms) and analyzed their data.
  - ✓ They have compared inverting and non-inverting configurations, highlighting key differences and applications.
  - ✓ Finally, they document their findings in a report, explaining the operation principles, observed performance, and practical applications of each amplifier type.

- **CO – PO / PSO mapping:**

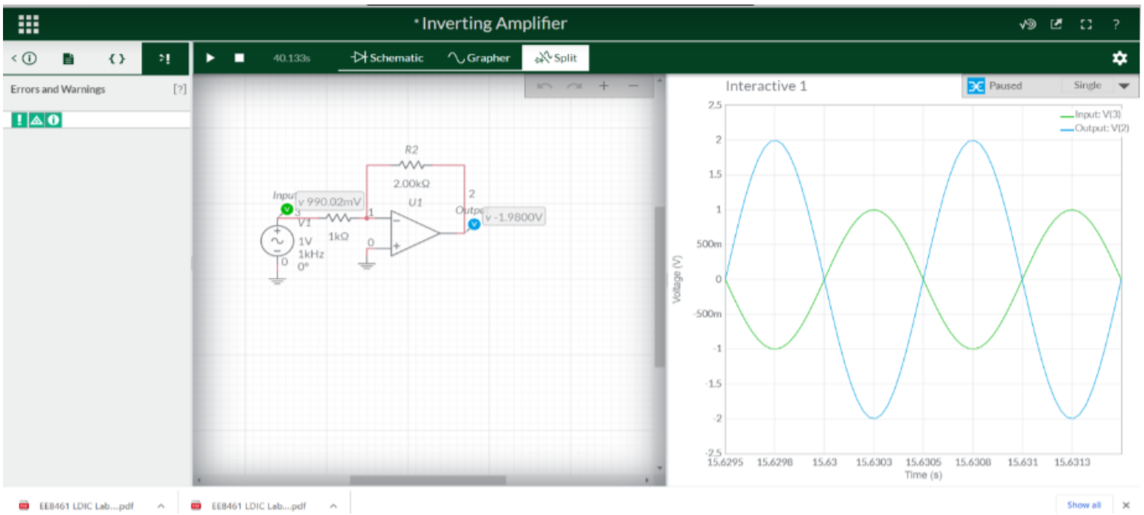
CO	PO1	PO2	PO3	PO5	PO9	PO12	PSO1	PSO2
CO2	2	1	1	1	1	1	1	1

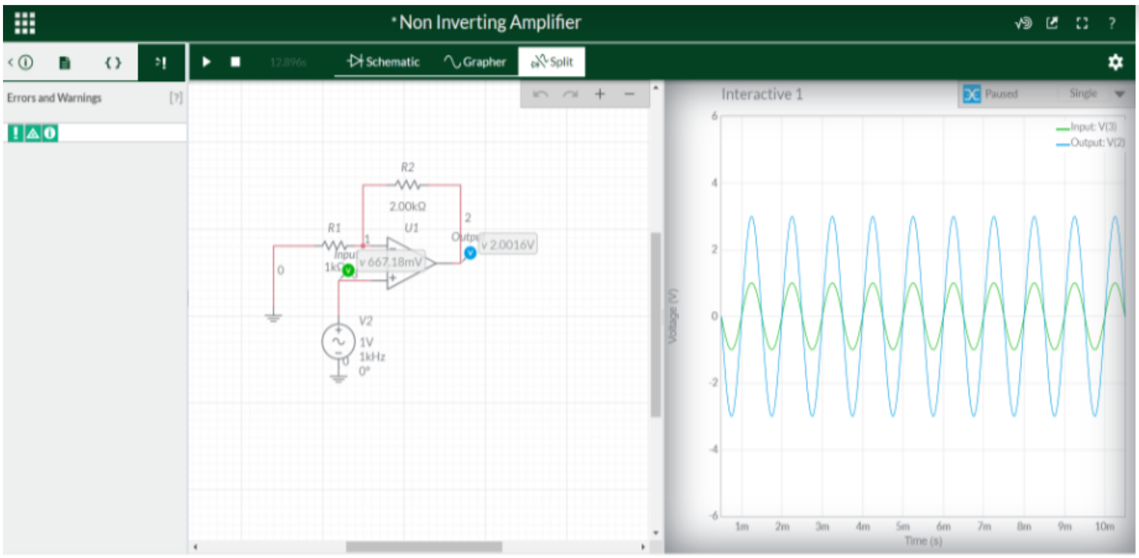
(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO5	PSO1
	1	1
Justification for correlation	The students are using the modern simulation software. Hence it is mapped with 1	The students are simulating the electronic systems for understanding the concepts. So, it is mapped with 1.

- **Images / Screenshot of the practice:**





- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - ✓ Students found the lab simulation activity engaging and insightful.
    - ✓ Students expressed positive feedback regarding the lab simulations. They enjoyed the interactive and hands-on approach, which made the concepts of inverting and non-inverting amplifiers clearer and more tangible.
    - ✓ Many appreciated the opportunity to visualize amplifier behavior and manipulate circuit parameters in a virtual environment.
  - ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
    - ✓ Immediate results from the software helped students correlate theoretical concepts with practical outcomes.
    - ✓ Simulations provide a safe and controlled environment for students to experiment with circuit design and operation. They can "build" and "test" circuits without the risk of damaging expensive equipment or facing safety hazards.
    - ✓ Simulations provide visual feedback, allowing students to observe the behavior of the circuit in real-time. This can help them understand the relationship between input and output signals, gain insights into circuit behavior, and identify potential problems.
    - ✓ Simulations enable students to easily modify circuit parameters (e.g., resistor values, input signals) and observe the resulting changes in output. This iterative process fosters a deeper understanding of circuit design principles.
    - ✓ Simulations help students develop valuable skills such as circuit analysis, troubleshooting, and design optimization.
  - ***Challenges faced in implementation:***
    - ✓ Some students mentioned initial difficulties in using the simulation software but valued the practical understanding gained through experimentation.
    - ✓ Limited session time made it challenging to explore advanced configurations or more complex scenarios.
    - ✓ Occasional software glitches or compatibility issues can disrupt the learning process.
    - ✓ Understanding how to effectively use simulation tools alongside theoretical concepts can be daunting for some students.

References:

- ❖ David A. Bell, ‘Op-amp & Linear ICs’, Oxford, Third Edition, 2011
- ❖ D. Roy Choudhary, Sheil B. Jani, ‘Linear Integrated Circuits’, New Age, Fourth Edition, 2018.
- ❖ https://www.multisim.com/help/getting-started/
- ❖ https://neuron.eng.wayne.edu/ECE330/multisim_tutorial.pdf

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Department of Electrical and Electronics Engineering
Academic Year 2023 – 2024 Even Semester

Degree, Semester & Branch: IV Semester B.E EEE
Course Code & Title: EE3402 – Linear Integrated circuits
Name of the Faculty member (s): Dr. K. Karthikeyan

Innovative Practice Description

- **Unit / Topic: Unit III / ADC and DAC and its types**
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** TLO 10
- **Activity Chosen: One-minute paper**
- **Justification:**
  - ✓ The **One-Minute Paper** activity is ideal for the topic "Analog to Digital and Digital to Analog Converters" as it encourages students to reflect on key concepts, such as conversion principles and applications.
  - ✓ Students are encouraged to identify areas where they need further clarification.
  - ✓ This quick assessment identifies gaps in understanding, promotes concise thinking, and allows instructors to address misconceptions promptly.
- **Time Allotted for the Activity:** 15 minutes
- **Details of the Implementation:**
  - ✓ At the end of the session, I have asked the students to write down in one minute, the key difference between ADC and DAC, their working principles, or real-world applications.
  - ✓ They wrote concise responses within one minute.
  - ✓ I have collected the papers from the students. Quickly reviewed the papers to identify common questions, misconceptions, and areas where further clarification is needed.
  - ✓ I have addressed common questions and misconceptions in the next class.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO9	PO12	PSO1	PSO2
CO3	2	1	1	1	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO10**
	1
Justification for correlation	The students are articulating the contents and communicating it to the whole class. Hence it is mapped with 1

- **Images / Screenshot of the practice:**

- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***
    - ✓ Students found the One Minute Paper activity both engaging and effective.
    - ✓ They appreciated the opportunity to quickly consolidate their learning and felt it was a useful tool for identifying areas they didn't fully understand.
    - ✓ Some students mentioned that the exercise forced them to think critically about the material and helped reinforce their understanding of Digital to Analog Converters (DAC) and Analog to Digital Converters (ADC).
    - ✓ Some found it challenging to summarize their thoughts within a minute.
  - ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
    - ✓ Students actively reflected on their understanding, reinforcing key concepts like the differences and applications of DAC and ADC.

- ✓ The activity created a focused and interactive environment, prompting participation from all students.
- ✓ The student responses provided me real-time insights into student comprehension, allowing targeted interventions.
- ✓ Students practiced summarizing and identifying core ideas, enhancing analytical skills.
- ✓ Provided a platform for students to voice their questions and concerns.

❖ ***Challenges faced in implementation:***

- ✓ Some students struggled to articulate their understanding within the one-minute limit.
- ✓ Differences in prior knowledge led to varied quality in responses, requiring the instructor to address a range of understanding.
- ✓ Collecting and reviewing responses in a short time frame required efficient coordination.
- ✓ Differences in students' writing abilities can affect the quality of feedback received, with some students providing more detailed and insightful reflections than others.
- ✓ Ensuring that all students take the activity seriously and provide meaningful feedback can be challenging.

References:

❖ David A. Bell, 'Op-amp & Linear ICs', Oxford, Third Edition, 2011

❖ D. Roy Choudhary, Sheil B. Jani, 'Linear Integrated Circuits', New Age, Fourth Edition, 2018.

❖ https://www.analog.com/media/en/training-seminars/design-handbooks/microprocessor-systems-handbook/Chapter7.pdf

❖ https://www.rochester.edu/college/teaching/teaching-guidance/one-minute-paper.html.

❖ https://www.westernsydney.edu.au/__data/assets/pdf_file/0020/1133363/one_minute_paper_1.pdf.

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Course Code & Title: EE3402 – Linear Integrated Circuits
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Innovative Practice Description

- **Unit / Topic: Unit IV / PWM application of IC555 timer**
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO 11
- **Activity Chosen: Sage and Scribe**
- **Justification:**
  - ✓ The **Sage and Scribe** activity suits the topic "PWM Applications of IC 555 Timer" as it promotes active learning.
  - ✓ For PWM applications of the IC555 timer, this method ensures active engagement, reinforces key concepts, and encourages peer-to-peer teaching, which enhances retention and comprehension of this technical topic.
  - ✓ The "Sage" explains PWM concepts and IC 555 applications, enhancing verbal articulation, while the "Scribe" writes, reinforcing listening and comprehension.
  - ✓ Role-switching ensures equal engagement and a collaborative understanding of the topic.
- **Time Allotted for the Activity:** 15 minutes
- **Details of the Implementation:**
  - ✓ Students are paired. One student is designated the "Sage" (expert) and the other the "Scribe" (note-taker).
  - ✓ The Sage explains the application to the Scribe, covering concepts like duty cycle and frequency control.
  - ✓ The Scribe actively listens and records key points, diagrams, and any questions.
  - ✓ Roles are switched, and the process is repeated for a different PWM application.
  - ✓ Pairs can share their findings with the class, fostering a deeper understanding of PWM applications of the IC555 timer.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO9	PO12	PSO1	PSO2
CO4	2	1	1	1	1	1	2	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO10	PSO2
	1	**1**
Justification for correlation	The students are communicating with each other in a group. Hence it is mapped with 1	The students are learning IC555 applications in electronic systems. So, it is mapped with 1.

- **Images / Screenshot of the practice:**

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - ✓ Students generally appreciated the interactive nature of the sage and scribe activity.
    - ✓ The students reported better understanding and retention of PWM applications concept through peer explanation and collaboration.
    - ✓ Students appreciated the hands-on approach, finding it engaging and effective in reinforcing concepts. They particularly valued the collaborative discussions and the step-by-step demonstrations by the sage.
    - ✓ Some students noted that switching roles helped them grasp both perspectives.
  - ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- ✓ The scribe provides immediate feedback to the sage, asking questions and seeking clarifications, which reinforces understanding and corrects misconceptions in real-time.
- ✓ Peer discussions fostered teamwork and improved communication skills.
- ✓ The scribe role taught students to record processes systematically, which is crucial for technical fields.

❖ ***Challenges faced in implementation:***

- ✓ Groups often needed more time to complete the activity, especially when concepts were new or complex.
- ✓ Students initially struggled to understand their roles as sage or scribe, leading to inefficiencies.
- ✓ Some students felt the pace was too fast during the sage's explanation, making it hard to keep up.
- ✓ A few also struggled with balancing their roles in group discussions and completing the task.
- ✓ Variations in students' prior knowledge and confidence levels can affect the effectiveness of the activity. More advanced students may need to be paired carefully with those requiring more support.

References:

- ❖ David A. Bell, 'Op-amp & Linear ICs', Oxford, Third Edition, 2011
- ❖ D. Roy Choudhary, Sheil B. Jani, 'Linear Integrated Circuits', New Age, Fourth Edition, 2018.
- ❖ https://www.ahanet.org/sagescribe
- ❖ https://krejcicreations.com/4-simple-and-effective-engagement-strategies/

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Innovative Practice Description

- **Unit / Topic: Unit V / Switching regulator, Switched Mode Power Supply (SMPS)**
- **Course Outcome:** CO 5
- **Topic Learning Outcome:** TLO 17
- **Activity Chosen: Think-Pair-Share**
- **Justification:**
  Think-Pair-Share is an effective method for this topic because:
  - ✓ It allows students to initially grapple with the complexities of SMPS independently, fostering deeper understanding.
  - ✓ Partner discussions encourage students to articulate their thoughts, clarify misconceptions, and learn from diverse perspectives.
  - ✓ The sharing phase brings valuable insights to the entire class, enriching the overall understanding and promoting a collaborative learning environment.
- **Time Allotted for the Activity:** 15 minutes
- **Details of the Implementation:**
  - ✓ Think: Students individually reflect on SMPS working principles and advantages for 5 minutes.
  - ✓ Pair: In pairs, they discuss their understanding, clarifying doubts and exchanging insights and identify applications of SMPS for 5 minutes.
  - ✓ Share: Each pair shares insights with the class, fostering collaborative learning.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO9	PO12	PSO1	PSO2
CO5	2	1	1	1	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO10	PSO1
	1	**1**
Justification for correlation	The students are discussing with each other in a group. Hence it is mapped with 1	The students are discussing the SMPS for engineering applications. Hence it is mapped with 1

- **Images / Screenshot of the practice:**

- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***
    - ✓ It helped the students to understand the material better when they had to explain it to someone else.
    - ✓ The students enjoyed discussing the SMPS with the team. They learned from each other.
    - ✓ The activity made the class more engaging and less passive.
  - ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
    - ✓ Students actively engage with the material through individual research and peer discussion.

- ✓ Deepened understanding of complex concepts through explanation and clarification.
- ✓ Fostered teamwork and communication skills through partner discussions.
- ✓ Enhanced ability to articulate and defend their ideas.

❖ ***Challenges faced in implementation:***

- ✓ Ensuring sufficient time for each phase (Think, Pair, Share) can be challenging.
- ✓ Some students may not have been prepared for the discussion.
- ✓ It can be challenging to ensure equal participation from all students in the pairs.
- ✓ Some students may feel uncomfortable sharing their ideas in front of the class.

References:

❖ David A. Bell, ‘Op-amp & Linear ICs’, Oxford, Third Edition, 2011

❖ D. Roy Choudhary, Sheil B. Jani, ‘Linear Integrated Circuits’, New Age, Fourth Edition, 2018.

❖ https://lth.engineering.asu.edu/2021/09/think-pair-share/

❖ https://www.kent.edu/ctl/think-pair-share

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